# Making sense of LLM outputs

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**Abstract**

Large language models generate fluent outputs, but understanding quality, reliability, and uncertainty remains difficult. This paper presents a simple framework for analyzing LLM outputs using semantic consistency, calibration-oriented confidence proxies, and task-grounded error taxonomies.

## 1 Introduction

Modern LLMs can solve a broad range of tasks, but evaluation is often reduced to aggregate benchmark scores. In many practical settings, users need to know *why* a response appears correct, *when* it fails, and *how* confident they should be in it.

This paper studies methods for making LLM outputs more interpretable and trustworthy. We focus on lightweight, model-agnostic analyses that can be applied post hoc to generated text.

## 2 Related Work

Prior work has examined hallucination detection, calibration, and response ranking by surrogate objectives [1]. We build on this literature and emphasize practical diagnostics that are easy to implement and compare.

## 3 Method

Let $x$ be an input prompt and $y$ be a generated response. We define a composite diagnostic score:

$$S(x, y) = \alpha C_{\text{sem}}(x, y) + \beta C_{\text{self}}(y) - \gamma E(y), \tag{1}$$

where $C_{\text{sem}}$ is semantic alignment to the task intent, $C_{\text{self}}$ measures self-consistency across sampled variants, and $E$ is a structured error penalty.

## 4 Experiments

We evaluate on representative QA and summarization tasks. For each task, we report:

- exact-match or task-specific quality,
- consistency across sampling runs,
- human-labeled error categories.

# 5 Results and Discussion

Preliminary findings suggest that combining consistency and error taxonomies improves detection of brittle outputs compared with single-score evaluation.

# 6 Conclusion

A structured view of LLM outputs can complement benchmark-centric evaluation and help practitioners reason about model behavior in deployment contexts.

# References

[1] Unknown. Ranking llms by compression. *arXiv preprint arXiv:2406.14171*, 2024.